BIRDB: A Secure and Efficient Encrypted Database using Compact Encrypted Search

Anonymous Submission #59

***Abstract*—The growing concerns about the confidentiality of data outsourced to untrusted cloud storage have prompted notable progress in *Encrypted Databases* (EDB). EDBs store encrypted data and allow for searches as traditional insecure databases while hiding the metadata of access patterns during searches. However, achieving both efficient encrypted search and metadata protection remains a challenge. Some EDBs construct cryptographic indexes on property-preserving encrypted data to enable logarithmic searches, but their performance downgrades as the index size substantially increases and also disregard metadata protection. Other approaches resort to scanning the entire dataset to hide metadata, leading to significant overhead during encrypted searches.**

We present BIRDB, the first efficient and compact EDB with metadata protection. At the core of BIRDB lies a new *boolean information retrieval* technique that structures the database in a logarithmic-complexity search data structure (index) and performs the search obliviously over the index. The boolean nature of the index provides two crucial advantages: (1) lossless index compression through linear approximation and (2) seamless integration with efficient private information retrieval protocols. This integration allows computationally expensive query-independent computations to be shifted to an offline phase, leading to sublinear computation during the query-specific online phase, all while ensuring metadata protection. Extensive evaluations show that BIRDB reduced up to 40.7% latency and achieved up to 45x higher throughput and provided robust metadata protection compared to existing notable approaches.

I. INTRODUCTION

The popularity of cloud computing has driven widespread adoption of cloud storage services such as DropBox [30] and Onedrive [79]. However, these services have raised grave security concerns due to their processing of user data in plaintext. Recent incidents, including a data breach exposing the personal information of 237,000 US government employees [1], the leak of Samsung's software source code when the employees uploaded it to ChatGPT [7], and accusations against TikTok for compromising the data of over 150 million US citizens stored in the cloud [8], highlight the serious data security risks associated with cloud storage.

Recently, *Encrypted Databases* (EDBs) have emerged as a dependable solution for enhancing the security of cloud storage. EDBs store user data in an encrypted form without knowing the decryption keys, ensuring data confidentiality even against privileged database administrators and external attackers (i.e., *adversaries*). To enable efficient query processing on the encrypted data, EDBs require efficient and compact search indexes, comparable to the binary search tree (BST) used in traditional insecure databases [56]. Furthermore, EDBs must employ oblivious algorithms to protect search metadata, preventing adversaries from extracting sensitive information from index access footprints during searches [23], [32], [84].

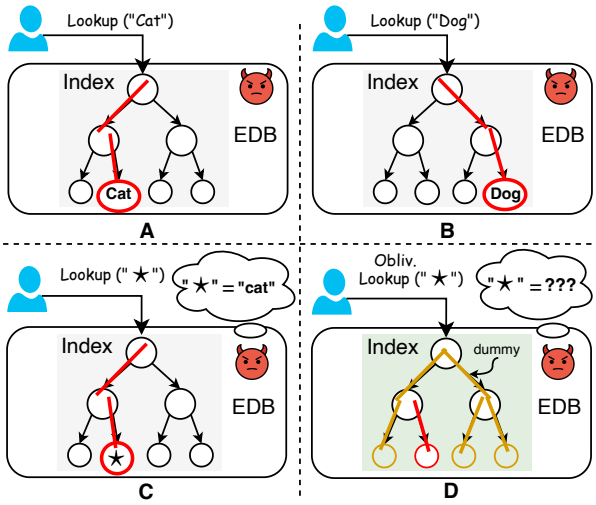


Fig. 1: Compared to an EDB that uses only end-to-end encryption shown in **(a)**∼**(c)**, a metadata-hiding EDB in **(d)** disguises intended index operations with dummy operations, preventing adversaries from learning the metadata (i.e., index access footprints during searches).

Figure 1 illustrates how an adversary can exploit search access patterns to extract plaintext information and emphasizes the advantages of metadata protection. Consider an EDB that employs a BST search index, responsible for mapping encrypted keywords to their corresponding satellite data such as addresses. The adversary passively monitors a series of plaintext queries and their associated index traversal paths (**A**∼**B**). By repeating this process, the adversary can deduce the keyword associated with each index path. Consequently, when a user submits a confidential query with an encrypted search keyword ('⋆' in **C**), the adversary can infer the keyword based on the specific index node accessed. Nevertheless, an EDB implementing metadata protection utilizes oblivious algorithms that include dummy accesses (as depicted in **D**), effectively concealing the actual search access patterns and mitigating the vulnerability associated with search pattern exposure.

To protect metadata and enable search functionality on encrypted data, a traditional approach is to use Oblivious RAM (ORAM) [31], [62], [73]. ORAM is typically structured as a BST and employs dummy accesses to cryptographically conceal the actual path traversed during searches [29], [35], [71]. However, despite the polylogarithmic asymptotic complexity of ORAM in relation to the size of the database, the existing ORAM schemes remain prohibitively expensive.

There are two primary reasons why ORAM is not suitable for our specific setting. Firstly, the **one-phase design** of ORAM requires repeatedly executing the entire set of oblivious algorithms for each online search [26], [62], [71]. Secondly,

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the cryptographic nature of ORAM makes it **uncompressible** due to the introduction of randomness during cryptographic operations. As a result, a giant ORAM-based index containing a substantial volume of data needs to be partially transferred from fast primary storage to secondary storage, even in industrial deployments [27], [33]. This reliance on costly I/O operations to secondary storage becomes a bottleneck and leads to severe performance degradation (proved in §VI-A).

As an alternative to ORAM, OO-PIR [53] proposes a two-phase private information retrieval (PIR) design. It preprocesses the database into multiple chunks, where each online query retrieves a random chunk to conceal the specific item being fetched. While OO-PIR moves costly oblivious operations (chunk division) to an offline phase, it still requires storing the entire set of chunks and linearly scanning each chunk during online queries. Another approach, HELibDB [24], utilizes Fermat's Little Theorem [78] to assess the equivalence of encrypted data, eliminating the need for index construction and achieving compactness. However, this direct evaluation necessitates linearly scanning a large amount of encrypted data, which hampers performance.

To enhance search efficiency and expedite the aforementioned metadata-hiding solutions, one approach is to utilize property-preserving encryption algorithms [17], [9] to construct cryptographic indexes on the encrypted data. These algorithms preserve the determinism and order of the encrypted data, thereby enabling the construction of BSTs with logarithmic search complexity. However, this cryptographic index is not suitable due to its incompatibility with the PIR process in existing metadata-hiding solutions [24], [53]. Additionally, the cryptographic index also suffers from the limitation of uncompressibility, further restricting its performance.

Overall, the challenge of striking a balance between efficiency (logarithmic search complexity) and compactness (compressible encrypted index) while ensuring metadata protection in an EDB remains an open problem. We believe that the root cause is the use of mutually isolated cryptographic primitives to address individual challenges, rather than seeking a unified solution that can rule all aspects.

Drawing inspiration from Boolean Information Retrieval protocols developed for document retrieval [11], [48], we made a key observation that the boolean form of encrypted data can provide significant *membership* information, which can assist in achieving all three desired properties simultaneously. Firstly, the boolean form enables a new encrypted search concept by replacing traditional numerical comparisons in BSTs with membership tests. Secondly, the boolean characteristic of the data allows for the use of linear approximation [45], [49], which compresses the encrypted indexes within constant space occupancy, overcoming the inherent uncompressibility of cryptographic indexes in prior work. Lastly, the boolean form of the data aligns well with the two-phase PIR process [39], [53], where the database is preprocessed, and users ultimately recover results using boolean operations.

This paper presents BIRDB[1], the first encrypted database with metadata protection that achieves efficiency and compactness. The key weapon of BIRDB is a new compressible

[1]BIRDB stands for **B**oolean **I**nformation **R**etrieval based E**DB**.

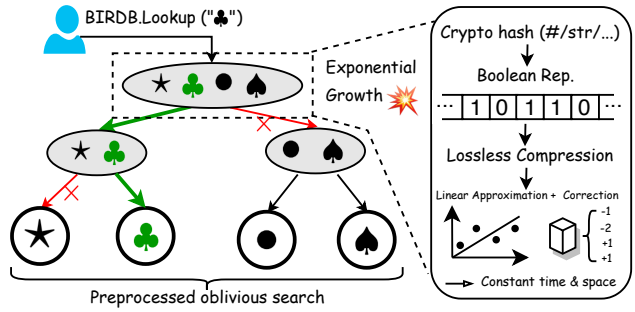


Fig. 2: BIRDB's fundamental concept revolves around leveraging boolean information to construct a searchable and compact index that incorporates an oblivious search protocol.

encrypted index called BINDEX. Unlike prior cryptographic approaches that build uncompressible indexes due to random noise introduced during encryption, BINDEX utilizes efficient one-way functions (OWF) such as SHA-256 [38]. These OWFs transform the data into irreversible ciphertexts in boolean form and then BIRDB organizes them in a membership-preserving data structure. The membership information is propagated from the leaf to the root, enabling logarithmic search, while the boolean nature of the index allows for compressibility.

Figure 2 illustrates the new encrypted search method employed by BINDEX, showcasing its construction and search process. During the index construction, each item in the database undergoes a transformation using OWFs, resulting in boolean-form cipher identifiers. These identifiers are positioned at the leaf layer of the data structure. BINDEX is then constructed by recursively combining these identifiers in a Merkle tree fashion, where each inner node represents the *union* of its child identifiers. This process ensures that the membership information of items is propagated up to the root of the tree. During encrypted searches, BINDEX employs a membership testing approach in each node to determine the appropriate path to traverse (§IV-A). This enables binary encrypted search by testing the membership of identifiers, mimicking the behavior of BSTs without the need for numerical comparisons on plaintext values.

A key performance challenges faced by BINDEX is its space efficiency. BINDEX needs to be compressed, and the need for compression is even more critical due to its unique leaf-to-root membership propagation. This membership propagation mechanism, which enables logarithmic-complexity encrypted searches, amplifies the space occupancy of BINDEX. The reason is that inner nodes in BINDEX are required to maintain the membership information of all their child nodes. As a result, the size of each inner node grows exponentially, leading to a significant increase in the overall size of the index and significantly degrading the performance of BINDEX.

To address the space efficiency challenge of BINDEX while maintaining its searchability, we draw inspiration from recent advancements in the field of linear approximation models in the database community [44], [45], [49]. Our key idea is to use *offsets* to represent each giant BINDEX node in boolean form. These offsets are then compressed by approximating the entries in the offset arrays using succinct linear models. This approach allows us to achieve compressed BINDEX nodes with constant space occupancy, as only a few slopes and intercepts in the models need to be stored (§IV-B). By employing this

compression technique in combination with the logarithmic search complexity of BINDEX, we can significantly reduce the performance penalty caused by the expanded index and enable BIRDB's queries to run at high performance without compromising data confidentiality.

Moreover, the boolean nature of BINDEX is compatible with and can be seamlessly integrated into two-phase PIR protocols. In practice, BINDEX is constructed and deployed on two servers: a master server and a replica server. The replica performs database preprocessing to generate auxiliary hints, which are then used by the master server during the online phase to provide efficient responses to user queries with sublinear computation and communication. This integration is seamless because the preprocessing step also relies on boolean operations (§IV-C). The integration between BINDEX and the PIR protocol is mutually beneficial: BINDEX accelerates the PIR process for encrypted searches, while the PIR protocol preserves the metadata confidentiality during searches.

We implemented BIRDB on OO-PIR's codebase [25]. BIRDB's modular architecture allows for independent deployment of index and storage optimizations, providing flexibility and transparency. To evaluate its performance, we conducted a comparative analysis of BIRDB against an uncompressed version (UC-BIRDB), HElibDB [24], and OO-PIR [53], which are two efficient metadata-hidng EDBs, using the same NoSQL workload [86]. Evaluation shows that:

• BIRDB is efficient: BIRDB achieved 5.9x lower latency and 45x higher throughput compared to HElibDB. It achieved 40.7% lower latency and 3.2x higher throughput compared to OO-PIR. The compressed BINDEX occupied approximately 1018x less space than UC-BIRDB, resulting in further reduced latency by 1.15x. These performance gains can be attributed to the low complexity and compactness (§VI-A).

• BIRDB is scalable: BIRDB's query latency scales logarithmically with an increasing size of the database (§VI-B).

• BIRDB is lossless: BINDEX successfully performed encrypted searches without any errors based on empirical evaluations and theoretical analysis presented in §IV-B.

Our main contribution is BINDEX, the first efficient and compact index for encrypted search. BIRDB makes the first effective attempt to use boolean information for constructing a searchable and compressable index, and seamlessly integrates with efficient two-phase PIR protocol that preprocesses database using boolean operations, in one unified system. BIRDB has potential to be deployed in many safety-critical applications, including privacy-preserving video streaming [63], [83], password manager [34], recommendation [40], [68], and anonymous messaging [42], [67]. BIRDB is open-sourced and its code is released on github.com/ndss24p59/BirDB.

II. BACKGROUND

In this section, we aim to highlight the importance of meeting three key properties - *searchability*, *compactness*, and *metadata protection* - in encrypted databases (compared in Table I). Understanding the significance of these properties will provide motivation for our design of BIRDB.

System	Data Confidentiality	Metadata Security	Sublinear Complexity	Compact Index	General Purpose
MySQL [56]	✗	✗	✓	✓	✓
CryptDB [64]	✓	✗	✓	✗	✓
HElibDB [24]	✓	✓	✗	✓	✓
Dory [27]	✓	✓	✗	✗	✗
OO-PIR [53]	✓	✓	✗	✗	✓
Titanium [53]	✓	✓	✗	✗	✓
BIRDB	✓	✓	✓	✓	✓

TABLE I: Characterization of representative EDBs.

A. Toward efficient server-side search

In response to growing security concerns about data leakage, enterprises like Keybase [5], PreVeil [6], SpiderOak [2], Sync [3], and Tresorit [4] have emerged, offering encrypted storage services to ensure user data confidentiality. These companies have expressed a need for server-side search functionality on encrypted data, in a way similar to conventional plaintext databases like MySQL [56]. However, none of these companies currently support server-side search, as their development efforts have been hindered by performance or the disclosure of search access patterns.

One straightforward yet inefficient way to enable search on encrypted data is to download the entire database. However, this approach is impractical due to the significant bandwidth consumption and client-side storage resources required. Thus although it provides the highest level of security, no existing industrial solution utilizes this approach.

To enable efficient server-side search on encrypted data, extensive academic research has explored the use of specific cryptographic tools, such as property-preserving encryption, as seen in the case of CryptDB [64]. The core concept behind this approach is to preserve the determinism and maintain the order of encrypted data, thereby allowing for the direct construction of an index (typically a BST [22]) on top of the encrypted data with logarithmic search complexity.

The need for compacting index. Despite extensive efforts to achieve efficient search performance, evaluations have shown that the actual performance often falls short of theoretical (i.e., logarithmic) expectations. The primary reason for this discrepancy is that the performance bottleneck does not lie in the binary search itself. Instead, it is the I/O operations to external storage that significantly degrade the overall performance. This is particularly true when dealing with substantial data volumes that cannot be indexed and fit within the limited main memory, compounded by the fact that encryption algorithms typically expand ciphertext size linearly [10], [74].

Therefore, in the context of EDB, achieving high performance necessitates not only sublinear (logarithmic) search complexity but also effective compression techniques. By ensuring sublinear search complexity, the search operations can be executed efficiently, while compression helps mitigate the impact of I/O operations and reduces the storage requirements.

B. Toward efficient metadata protection

Private Information Retrieval (PIR). The initial concept of Private Information Retrieval (PIR) was introduced by Chor et al. [57], allowing clients to retrieve specific objects from an untrusted server-managed database without revealing

the accessed object. PIR protocols aim to minimize communication costs, achieving sublinear complexity relative to the database size. Modern PIR deployment models include multi-server PIR [28], [47], [50], [52], where the database is replicated across non-colluding servers for enhanced fault tolerance. Alternatively, single-server PIR protocols [54], [55], [60] are computationally more demanding but avoid non-collusion assumptions through cryptographic hardness assumptions. These advancements provide a range of options for secure and private information retrieval, considering efficiency, security assumptions, and deployment requirements.

Oblivious RAM (ORAM) is a traditional PIR technique known for concealing access patterns during interactions with untrusted servers [61], [72], however, it is not suitable for our setting due to its impractical performance penalty. ORAM protocols achieve access pattern obfuscation through techniques like oblivious data shuffling and path encryption. Researchers have also proposed various ORAM variants like Titanium [23] and Dory [26] to improve performance and scalability while preserving data privacy [19], [37], [66], [70], [71]. However, ORAM-based techniques still suffer from the drawback of processing the entire database with linear computation for every query, resulting in significant performance penalties.

Two-phase PIR on the boolean information. Many research efforts have been put on improving the performance of ORAM by introducing preprocessing techniques [16], [25], [69], [85]. Preprocessing involves shifting the inevitable PIR computation to a query-independent offline phase and generating auxiliary data or hints along the way. This approach aims to make the computational cost of online query processing, with the assistance of these hints, sublinear to the database size.

Two recent PIR protocols, namely the Corrigan-Gibbs and Kogan scheme (CK) [25], and the Shi, Aqeel, Chandrasekaran, and Maggs scheme (SACM) [69], operate on the boolean information to enable a two-phase design. Notably, the CK protocol [25] presents a two-server PIR construction that stores $\mathcal{O}(\sqrt{n})$ bits at the client and achieves $\mathcal{O}(\sqrt{n})$ server computation for online queries. The fundamental idea behind CK is the reflexivity of XOR operations. The CK protocol divides the database into random chunks, and the user preserves a preprocessed hint $h = p \oplus s$, where p represents the desired value to be queried and s is obtained by querying a random chunk of the database. Through this approach, the user can ultimately obtain $p = h \oplus s$ using its hint and queried random result, with correctness guaranteed by the reflexivity property (i.e., $p \oplus s \oplus s = p$).

C. Opportunities

Based on the survey results discussed above, we have identified several opportunities to address the challenge of practical yet secure encrypted search:

• Traditional cryptographic approaches are not capable of simultaneously achieving both searchability and compactness. Hence, there is a strong need for an alternative approach that offers a more flexible and practical index design, even if it entails sacrificing some degree of confidentiality. This trade-off is crucial to ensure the compactness of the encrypted search system, making it efficient for real-world applications.

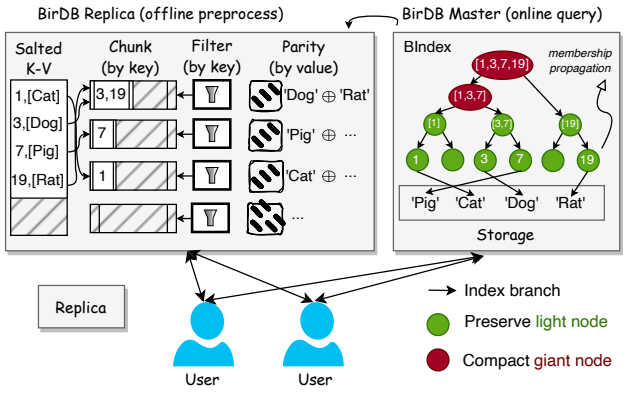


Fig. 3: BIRDB's architecture for searchable, compact, and metadata-hiding encrypted database built on the two-phase PIR.

• Boolean information has shown significant advancements in efficient metadata protection and holds great potential for implementation in an encrypted search index.

These opportunities serve as the basis for BIRDB's design.

III. Overview

A. System model

Same as [26], in BIRDB, the database items are divided into partitions, each managed by a different group of EDB servers. Multiple partitions can exist in a query system, and execution across partitions occurs in parallel.

Entities. For a single partition (depicted in Figure 3), we adopt the two-phase PIR model, specifically the CK protocol introduced by Corrigan-Gibbs and Kogan [25]. In our setup, a single partition consists of two servers deployed on the public cloud: an offline server (usually a replica) responsible for database preprocessing to generate runtime hints for PIR, and an online server (master) handling user queries. These servers are *semi-honest*, in the sense that they do not collude but have an interest in learning which objects the client is accessing from the database. Concretely, BIRDB's architecture for a single partition includes the following entities:

• **Users:** Multiple users interact with the BIRDB replica to obtain hints and send search queries (e.g., point lookup) to the BIRDB master for information retrieval. Each user needs to store a *salt*, a randomly generated 128-bit key, used for salting and recovering the original data during offline preprocessing to ensure confidentiality.

• **Master:** The BIRDB master processes user queries by searching BINDEX sublinearly and performing sublinear PIR computations. The master also ensures that the l replicas have a consistent view of the database state, and users know which servers to contact. In line with [26], we employ $l = 2$ in our implementation.

• **Replicas:** The BIRDB replica conducts database preprocessing to generate runtime *hints*, ensuring metadata-hiding during information retrieval while keeping the computation required on the master sublinear.

Similar to [26], BIRDB does not provide availability if servers refuse to provide services. However, it is important to note that cloud providers are typically incentivized to ensure

availability by either bringing the failed server back online or replacing it promptly [26], [43], [80].

Data model. To exemplify BIRDB's design, we employ the NoSQL key-value data model for its simplicity. However, it's important to note that BIRDB is highly extensible and can be applied to other data models transparently, including structured relational data models and unstructured key-value pairs. This is made possible by the unclustered architecture on which BIRDB is built, as discussed in [56], which decouples BIRDB's encrypted search index from the underlying storage.

Similar to previous works [53], [64], BIRDB replicates the database D across both the offline and online servers. The database consists of two confidential columns, each containing n keys or values (items). Each item has a size of b bits. Before sending values to the replicas, users encrypt them using their salts to ensure confidentiality during preprocessing. The preprocessed results, also known as hints, can later be recovered by the user using the respective salts. The correctness of this process is guaranteed by the reflexivity property of XOR on boolean operations, as we will explain in §IV.

In line with typical PIR systems [26], [53], the value column in database D is treated as an array, where $D[i]$ represents the i-th value in D. The notation $[x, y]$ is used to denote the set of consecutive integers from x to y, inclusive. For simplicity, when x = 1, the notation $[y]$ is used to represent the set $\{1, ..., y\}$.

PIR model (two-phase PIR [25])**.** BIRDB is built on the two-phase PIR which typically comprises four algorithms (`Prep`, `Query`, `Resp`, `Recov`). These algorithms are defined as follows in the setup of BIRDB over a database D of n items:

• `Prep`$(D) \rightarrow h$: this algorithm is executed by BIRDB replicas for each user. It takes the database D as input and outputs a hint h. The hint h consists of three parts: randomly partitioned chunks, filters for chunks, and parities for chunks.

• `Query`$(h, i) \rightarrow q_i$: this algorithm is executed by each user. It takes the hint h and the desired query index i as input and outputs a query q_i.

• `Resp`$(D, q_i) \rightarrow r_i$: this algorithm is executed by the BIRDB master. It takes a query q_i from users as input and outputs a response r_i.

• `Recov`$(h, r_i) \rightarrow d_i$: this algorithm is executed by each user. It takes as input a hint h from a replica and a response r_i from the master, and outputs the desired data object d_i.

These four algorithms work together to enable the metadata-hiding retrieval of specific data objects in BIRDB, ensuring that the access patterns and queried data remain concealed from the untrusted servers.

B. Threat model

In line with recent cloud-based EDBs [23], [26], [53], [64], we assume semi-honest database servers, in the sense that these servers do not collude but are interested in learning the plaintext data and the accessed items. Adversaries, denoted as $\mathcal{A}$, can include database administrators and privileged insiders who can perform passive attacks such as cipher analysis [18] and dictionary attack [20] to extract plaintext information. We rely on the security of one-way functions (OWFs) like SHA-256 [38], assuming that $\mathcal{A}$ is computationally incapable of reversing OWFs or compromising cryptographic encryption keys. These assumptions form the foundation for ensuring the confidentiality of both the data and metadata in BIRDB's encrypted data during encrypted search.

We assume that users are always honest and do not engage in collusion with any servers. While access control mechanisms [58] and blacklists [77] can be used to prevent corrupted clients, we do not consider them within the scope of this work. Additionally, similar to prior research [26], we assume that the underlying storage engine is maliciously secure. This ensures that servers can retrieve the correct data from the underlying storage by detecting rollback and fork attacks, which have been extensively studied in previous storage works [21], [75], [46].

Out-of-scope attacks. We make the assumption that servers do not engage in active attacks such as modifying, dropping, or replaying queries, and that they adhere to BIRDB's protocol. Mitigations exist to authenticate queries and ensure query integrity on plaintext data under active attacks, but addressing integrity under active attacks on encrypted data remains an open problem that we leave for future work [82], [88]. Furthermore, we assume a fixed size for items, which eliminates size leakage. It is challenging to prevent size leakage efficiently without extensive padding to the largest file size (which is costly) or relying on trusted hardware (which requires trust in a third-party vendor). However, mitigations such as partial padding and delayed accessing, where different file chunks are downloaded at different times, can be employed [59], [76].

BIRDB's guarantees. BIRDB's confidentiality guarantee ensures that all index keys are irreversible, in the sense that if $\mathcal{A}$ is a computationally bounded adversary, $\mathcal{A}$ is infeasible to reverse the encrypted index keys and retrieve the original records in plaintext. These guarantees are facilitated by BIRDB's index, which effectively achieves high-performance encrypted search with logarithmic complexity and compactness. Formally, we have the following theorem (proved in §V) that captures BIRDB's security:

Theorem 1. *Given that the one-way functions (OWFs) used in index construction is mathematically irreversible, and assuming that the ϵ-approximate linear models and the two-phase PIR scheme of Corrigan-Gibbs and Kogan [25] are secure against passive attacks, the* BIRDB *encrypted database is secure in the presence of a computationally bounded adversary.*

C. System goals and approaches

At a high level, our goal is to achieve searchability, compactness, and metadata protection for encrypted search, while maintaining the confidentiality of both data and metadata. To accomplish this, our key weapon is a new boolean information retrieval (BIR) technique that allows us to achieve all of these objectives simultaneously. Below, we provide a brief overview of the approaches to achieve these desirable features in EDBs.

Goal 1: Logarithmic encrypted search. Enabling search functionality on encrypted data is crucial for efficient retrieval of specific items without the requirement of decrypting the entire dataset. To address this challenge, our key idea is to replace conventional numerical comparison operations at

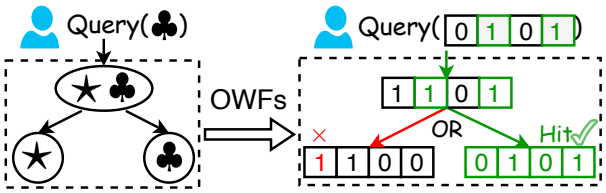


Fig. 4: BIRDB employs boolean information for logarithmic encrypted search by testing membership in index branches and finally comparing every bit at leaf nodes.

index branches with secure boolean membership tests. By incorporating secure boolean information into an orchestrated searchable data structure, we can enable encrypted searches with a logarithmic time complexity.

Figure 4 exemplifies our encrypted search method named BINDEX. This approach involves converting database items into boolean information using cryptographic one-way functions (OWFs), such as SHA-256. Drawing inspiration from the Bloom Filter [13], BINDEX employs a membership-preserving data structure to maintain the membership of the OWF-transformed items, which are represented as binary strings (e.g., '⋆' is represented as '1100'). These binary strings serve as leaf nodes positioned at the lowest index layer, while the inner nodes at higher levels are the *unions* of their child nodes. Each inner node performs an *OR* gate computation on the boolean entries of its child nodes to generate its own boolean information. To search for a specific item, BINDEX compares the '1' bits of the item at each index branch, starting from the root, to ensure a match for every '1' bit. At the leaf node, BINDEX compares each entry, including '0' bits, to ensure an exact match with the queried item.

Goal 2: Compact index with constant overhead. Although achieving logarithmic search complexity (**Goal 1**) is important, it is not sufficient to ensure high performance. This is because as the inner node size of BINDEX grows exponentially when reaching the index root, it becomes challenging to store a large number of index nodes in primary storage. As a result, more nodes are shifted to the secondary storage, which incurs costly I/O operations and hampers performance.

Figure 5 illustrates our approach to compact BINDEX and achieve constant query time and space occupancy. The core idea is that BINDEX's membership information only needs to store the positions of the '1' bits, while the numerous

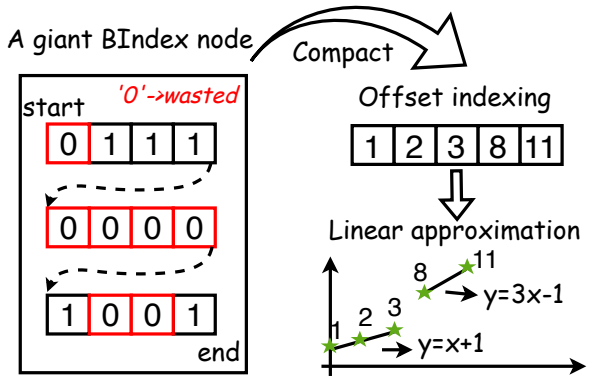


Fig. 5: Based on the boolean information, BIRDB compacts giant index nodes to optimize I/O efficiency and achieve high performance.

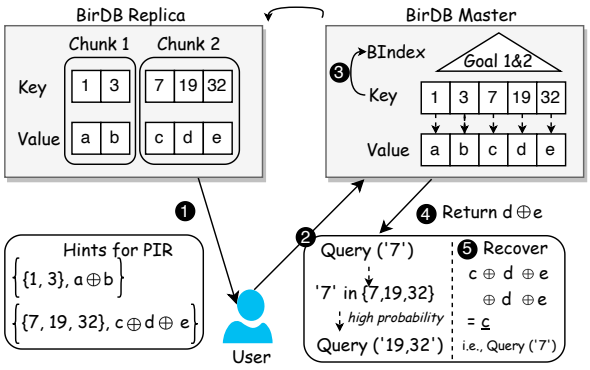


Fig. 6: The third goal of metadata-hiding in BIRDB is achieved through the utilization of two-phase PIR. BINDEX, which accomplishes **Goals 1 and 2**, accelerates the PIR process, while PIR ensures the metadata confidentiality of BINDEX.

'0' bits are left irrelevant. Taking inspiration from the recent new family of linear approximation model [44], [45], [49], we propose to use *offsets* to represent the boolean information (specifically, the positions of '1' bits). We then compact these offsets using piece-wise linear functions that map the position in the array to the corresponding offset [33]. For instance, we can utilize two linear functions, defined by only two slopes and intercepts, to represent the offset array belonging to a BINDEX node in Figure 5. Consequently, the implementation of this compacted structure requires constant time and space, regardless of the total number of stored items.

Goal 3: End-to-end metadata protection. To effectively conceal metadata and eliminate the leakage of search access patterns, we implement BIRDB using the CK protocol introduced by Corrigan-Gibbs and Kogan [25]. The CK protocol is a cutting-edge two-phase PIR protocol that offloads query-independent computations to a preprocessing phase, resulting in sublinear computation for online queries.

Figure 6 demonstrates the integration with two-phase PIR discussed in §III-B. Specifically, the process consists of a preprocessing phase and a query phase. During the preprocessing phase, the BIRDB replica divides the database into random chunks. For each chunk of keys, it computes the parity of the corresponding values and builds filters for each chunk (❶). In the query phase, the user searches for their desired query item by testing its membership in each chunk's filter. Once the target chunk is found, the user deletes the intended query item (❷). The BIRDB master then searches the database using BINDEX based on the query items and computes the parity of the searched values (❸ and ❹). Finally, the user recovers the result by XORing the returned parity with the parity of the original chunk (❺). The correctness of this process is ensured by the reflexivity of XOR on boolean operations; BIRDB accelerates the PIR process using BINDEX (§IV-C).

Overall, by utilizing one core concept of boolean information retrieval, we construct a search index with logarithmic complexity using OWFs. Furthermore, we compact the index by employing linear approximation techniques. Finally, we ensure the concealment of search metadata by leveraging the reflexivity of XOR on boolean operations. Hence, we address all three essential properties: searchability, compactness, and metadata-hiding, in a unified manner.

IV. Protocol Description

A. Desiging an encrypted search index

We first establish the foundation for constructing BIRDB's encrypted search index and subsequently delve into the details of index construction and search methods.

Preliminaries. The underlying searchable data structure in BIRDB is based on the Bloom filter [13], a compact data structure designed for efficient membership testing. The Bloom filter operates by utilizing a single set to store the hash values of all items. Specifically, a Bloom filter B is represented by an m-bit array and employs k independent and uniformly distributed OWFs, i.e., cryptographic hash functions $\{H_k\}$. Each hash function maps an item to an index in the Bloom filter array. To insert an item x into the filter, we compute its hash using $\{H_k\}$ and set all the bits at the corresponding indices in B to 1. For testing whether an item y is a member of the set, we again hash y using $\{H_k\}$. If any of the bits at the corresponding indices in B are 0, then y is definitely not in the set (i.e., zero false negatives). Conversely, if all the bits at the corresponding indices in B are 1, y is considered to have a high probability of being in the set, albeit with a small configurable false positive rate [13].

Challenges and opportunities. As mentioned above, traditional Bloom filters offer valuable information regarding the *membership* of data for tests. In addition to this, we have observed two key characteristics of Bloom filters that make them promising for encrypted search applications. Firstly, a Bloom filter is *monolithic* in nature, meaning that all items are assigned to a single boolean set. This characteristic allows for efficient item membership tests related to data existence. Secondly, a Bloom filter ensures data *confidentiality* by utilizing one-way functions (OWFs) to transform original plaintexts into hash values of items. The use of OWFs makes the Bloom filter computationally irreversible, effectively hiding the original plaintexts. This security property relies on the assumption of the one-wayness of cryptographic hash functions, which has been extensively studied in prior research [13].

However, despite the valuable membership property and data confidentiality offered by a Bloom filter, the monolithic nature limits its functionality to supporting only membership tests by indicating whether an item (hash) is present in a set. Consequently, Bloom filters alone are not capable of functioning as an index for efficiently searching encrypted data. We observe two significant benefits of using Bloom filters for encrypted search:

• **Security for indexing**. In encrypted search, users do not require the encrypted index keys to be decryptable, as they are used solely for searching corresponding values. Therefore, the utilization of one-way functions (OWFs) to efficiently transform plaintexts into computationally irreversible ciphertexts is sufficient. OWFs provide the necessary security for the index keys without the need for decryption.

• **Compactness**. The boolean nature of Bloom filters allows for the application of cutting-edge compression techniques specifically designed for boolean data. Leveraging these techniques can lead to significant compression gains in the encrypted search index. This is particularly advantageous as traditional encrypted search indexes, such as Order-Preserving Encryption (OPE)-based indexes, are not easily compressible [64], [26]. By reducing the index size through compression, the encrypted search system can benefit from improved performance by avoiding the drawbacks associated with large index sizes and potential performance penalties. We introduce the details of BIRDB compression in §IV-B.

Algorithm 1: Minifilter-based encrypted search

```
1  Function PointSearch(root, key) do
2      if isLeaf(root) then
           ▽ Every corresponding bit should match
3          if key.rft = root.rft        ▷ Minifilter comparison
4          then
5              return root
       ▽ In inner nodes, every '1' bit should match
6      if isInner(root) then
7          if left_minifilter.contain(key) then
8              return PointSearch(root.left, key)
9          if right_minifilter.contain(key) then
10             return PointSearch(root.right, key)
11         return NULL                  ▷ Key is non-existent
12 Function RangeSearch(root, [key1, key2]) do
13     left ← PointSearch(root, key1)
14     right ← PointSearch(root, key2)
15     while left.next != right do
16         value[] ← left.next      ▷ Retrieve sorted leaf nodes
17     return value
```

Approach. To construct the encrypted search index, we modify the traditional monolithic Bloom filter by decomposing it into a collection of *minifilters*. Each minifilter is responsible for storing the hash (in bits) of a single item. In other words, a mini-filter represents a Bloom filter dedicated to an individual data item, where only one item (hash) is assigned to the set.

The minifilters are recursively organized to form a searchable index known as BINDEX. These minifilters have a fixed array size m, and are arranged in a Merkle-tree structure. Each minifilter is stored at a leaf node of the tree, representing an index key. The index keys, such as `"1100"` for record '⋆' in Figure 4, are sorted based on their plaintext key order. The values of the inner nodes, which include all nodes except the leaf nodes, are determined through an element-wise OR gate computation on the minifilters of their child nodes. Let's consider an inner node with l child nodes, denoted as Cn. For each bit (array entry) in the minifilter of the inner node, its value is computed as follows: $Sub_{in}[i] = Cn_1[i]$ *or ...* $or\ Cn_l[i]$, for all i in the range $[0, m]$. Note that, in the construction of BINDEX, only the leaf nodes in the minifilter tree store actual values, such as record addresses. The inner nodes serve the primary purpose of indexing, facilitating tree traversal for efficient search operations.

Design rationale. The searchability of BINDEX is achieved through the membership-preserving property of the minifilters. Specifically, when any entry in a child node's minifilter is set to `"1"`, the OR gate computation guarantees that the corresponding entry in its parent node's minifilter will also contain a `"1"` bit. For example, in Figure 4, the first entry of the minifilter for '⋆' is `"1"` while the first entry of the minifilter for '♣' is `"0"`. As a result, the first entry of their parent node's minifilter is computed as $1\ OR\ 0 = 1$.

The BINDEX design ensures that if an item exists in a minifilter node, it must also exist in its parent node. In other words, the membership information is propagated to higher-

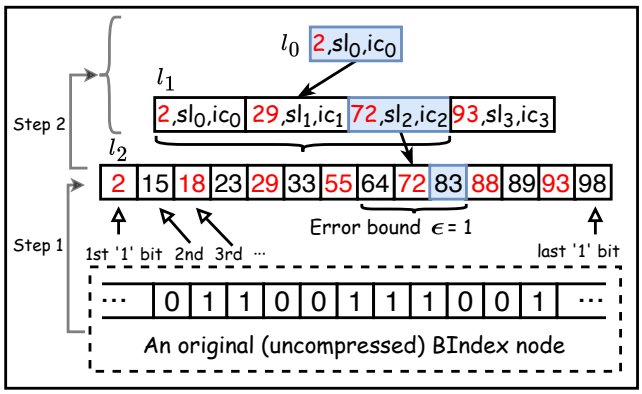


Fig. 7: The construction and a running example that show our compact design for BINDEX. **Step 1** extracts an original BINDEX node into offset representation wherein the ith entry is the position of the ith '1' bit. In **Step 2**, we uses the Piece-wise Linear Approximation model [33] to compact and search the extracted offset array.

level index nodes. This novel observation, utilized by BINDEX, allows for efficient top-down logarithmic tree traversal based on the membership information. Furthermore, the irreversible confidentiality provided by OWFs directly carries over to the construction of minifilters. By testing the membership of query items in the minifilter nodes, BINDEX eliminates the need for plaintext comparisons typically found in traditional binary search trees (BSTs).

Using BINDEX for encrypted searches. Algorithm 1 outlines the process of an encrypted point query through BINDEX, which involves generating a minifilter for the query item and traversing BINDEX to locate the corresponding value. BIRDB performs minifilter comparisons at each inner node to verify the membership of each `"1"` bit in the query item's minifilter. The traversal continues until reaching the leaf nodes, where the match of every bit confirms the presence of a unique index key.

The point search method can be extended to handle range queries. By transforming the range boundaries into minifilters, BINDEX locates two matched leaf nodes using the same method as point queries. All values between these leaf nodes are then returned, akin to the retrieval process in a B$^+$-tree.

B. Compacting BINDEX *for efficient encrypted search*

A core index challenge: size and cost. Despite BINDEX's efficient tree-based index structure, which facilitates logarithmic search complexity, achieving a low search complexity alone is insufficient for achieving high-performance encrypted search. The reason is two fold. Firstly, as the total number of items increases, each BINDEX node experiences exponential growth in membership information storage due to the membership propagation design (§IV-A). Consequently, the index size significantly expands, adversely affecting search efficiency. Secondly, even in a static setting, the false positive issue, attributed to the fundamental nature of Bloom filters, further amplifies the size of each minifilter node.

To illustrate, let's consider a database with 2^{20} static items and the user aims for a low false positive rate of 1×10^{-3}, same as the configuration in [81], each minifilter in BINDEX would need to be approximately 1.79MB in size using the formula in Bloom filter theory [13]. Consequently, the entire BINDEX would require a substantial amount of storage space,

Algorithm 2: Encrypted search on a compressed minifilter

Input parameter:
  A ▷ Sorted offset array of a minifilter node
  ϵ ▷ Error bound of PGM (typically $\epsilon = 1$)
  $levels$ ▷ Level of compacted BINDEX
  k ▷ The query item for search
Function `Query`$(k, A, \epsilon, levels)$ **do**
  $pos \leftarrow f_r(k), where\ r \leftarrow levels[0][0]$
  while $i \leq size(levels)$ **do**
    $lo \leftarrow max\{pos - \epsilon, 0\}$
    $hi \leftarrow min\{pos + \epsilon, size(levels)\}$
    $s \leftarrow []$
    while $s' \in levels[i][lo, hi]$ **do**
      if $s'.key \leq k$ **then**
        $s \leftarrow s'$
    $t \leftarrow right\ most\ segment\ of\ s$
    $pos \leftarrow \lfloor min\{f_s(k), f_t(t.key)\}\rfloor$
  $lo \leftarrow max\{pos - \epsilon, 0\}$
  $hi \leftarrow min\{pos + \epsilon, size(A)\}$
  $return\ binary\ search\ for\ k\ in\ A[lo, hi]$

around 3.57TB, which is unaffordable. The significant increase in space occupancy compared to a plaintext database index results in a notable performance decline due to expensive I/O operations involving secondary storage such as disks.

Facing this challenge, our contention is that relying solely on achieving theoretical logarithmic search complexity is insufficient for achieving high-performance encrypted search, even though it represents a significant improvement over linear search methods [24], [27]. Instead, an encrypted search index should adopt a compact design to ensure that the overall query performance remains manageable and cost-effective.

The compact design. As depicted in Figure 7, during index construction, BINDEX preserves light index nodes and compacts giant index nodes that exceed a user-defined memory budget B using two steps below.

• **Step 1 (From bits to offsets).** Recall that in BINDEX, each index node (i.e., minifilter) is represented as a one-item Bloom filter (§IV-A), which is a bit array consisting of only '`0`' and '`1`' bits. However, only the '`1`' bits contain informative data since membership tests only involve checking if all '`1`' bits are present in a filter. Thus in **step 1**, we extract the positions of all '`1`' bits from the filter, creating a new offset array. This extraction process allows us to discard unnecessary bits and avoid wasting primary storage resources.

• **Step 2 (Linear Approximation using PGM [33]).** Although **step 1** reduces the size of giant BINDEX nodes, the size still increases exponentially with the total number of records n. Fortunately, we find that the cutting-edge compression technique known as Piece-wise Linear Approximation, particularly the PGM [33], is highly suitable for further compressing the extracted offset array in our specific setting.

Specifically, the PGM is a data structure that addresses the linear approximation problem in a multiset of sorted keys within an array A, drawn from a universe U. PGM is parametric with an integer ϵ and employs piece-wise linear approximation models to learn a mapping that returns a position for a given key $k \in U$, which is within ϵ distance from the correct position in A. The fundamental concept behind PGM is that a single linear model, or *segment*, may not be sufficient to ϵ-approximate the positions of all keys in U. Therefore, PGM

employs a sequence of segments, each requiring constant space (two floats and one key) and constant query time to provide the ϵ-approximate position of k in A. To construct each segment, PGM reduces the piece-wise linear ϵ-approximation problem to constructing the convex hull of a set of points using a streaming algorithm. This algorithm operates in linear time and space, generating the minimum number of ϵ-approximate segments. We refer to [36], [51], [65] for more on the convex algorithm.

Once the prepared segments, such as the two linear functions shown in Figure 5, are available, the next step is to identify the ϵ-approximate segment responsible for estimating the approximate position of a query item. To achieve this, PGM begins with the sequence constructed over the entire input array A. It then extracts the first key (indicated by red keys in Figure 7) from A that falls within the coverage of each segment. Subsequently, another linear model is constructed using this reduced set of keys. This recursive process continues until convergence to a single segment. Following the recursive construction, the ϵ-approximate positions returned by the segments can be converted into exact positions through a binary search within a range of $\pm\epsilon$ keys in A. This binary search operation has a time complexity logarithmic to ϵ, regardless of the total number of records in BINDEX's offset array.

Running example. Figure 7 provides a running example of searching a compressed BINDEX node using Algorithm 2. Each accessed node in the search path is highlighted in blue. We assume the compressed BINDEX has an error bound $\epsilon = 1$ and the key being searched is $k = 83$.

The search begins from the root segment $s' = l_0[0][0]$. The next position, $\lfloor fs'(k) \rfloor = \lfloor k * sl_0 + ic_0 \rfloor$, is computed to be 1 for the next level. The search then proceeds to locate k within the range $[1-\epsilon, 1+\epsilon]$ in l_1, considering the keys [2, 29, 72]. It is determined that the next segment for the search is $s'' = l_1[2]$ because $k > 72$. Finally, the position for the next (leaf) level, l_2, which represents the extracted offset array, is computed by evaluating $\lfloor fs''(k) \rfloor = 8$. Consequently, a binary search is performed to search for k within the range $[8-\epsilon, 8+\epsilon]$ in $l2$. Ultimately, it is found that k is located at position 9 since $A[9]$ = 83.

In the compacted BINDEX, Algorithm 2 functions as the internal search mechanism for `minifilter.contain(`k`)` in Algorithm 1. Combined, they constitute the complete end-to-end encrypted search algorithm of BIRDB.

C. Seamless integrating BINDEX with PIR

Our metadata-hiding solution builds on the two-phase PIR scheme of Corrigan-Gibbs and Kogan [25]. We start by describing the CK protocol and then we decribe our approach to seamlessly integrate BINDEX with it.

Overview of the CK protocol. As described in §II-B, the CK protocol comprises two phases: an offline preprocessing phase conducted on BIRDB's replica, and an online query processing phase involving BIRDB's master. The offline preprocessing phase is independent of the specific item that the user wants to query and involves the `Prep` step. This phase is performed on BIRDB's replica and includes the necessary preparations for the PIR search process. In the online query processing phase, which depends on the specific item being queried, the interactions occur between BIRDB's master (who has no knowledge of the preprocessed hints generated by the replica) and the user. This phase consists of the `Query`, `Resp`, and `Recov` steps, where the user submits the query, the master provides the corresponding response, and the necessary recovery actions are taken.

- `Prep(`D`)` $\rightarrow h$. The replica divides all index keys of the database items into $\sqrt{n}\log n$ independent random sets, denoted as S_1, ..., $S_{\sqrt{n}\log n}$, where each set contains $\sqrt{n}$ keys sampled uniformly at random. For each set S_i, the replica computes its parity, P_i, where $p_i = \oplus_{t \in S_i} D[t]$ for $i \in [1, \sqrt{n}\log n]$. Each parity is uniformly padded to b bits. In short, a parity p_i for set s_i is the XOR of all items in D referenced by the keys in S_i. The replica sends the hint $h = \{(S_1, P_1), ..., (S_{\sqrt{n}\log n}, P_{\sqrt{n}\log n})\}$ to the user.
- `Query(`h, i`)` $\rightarrow q_i$. The user generates a query for retrieving an item with index key i (i.e., D[i]). The user selects a set S from the hint h such that $i \in S$ (the corresponding parity is P_S). The user then removes i from set S with a high probability (i.e., $1 - \frac{\sqrt{n}-1}{n}$); otherwise, it removes a randomly chosen key from $S \backslash \{i\}$. The query q_i is transformed to the resulting set S^*, which has a size of $\sqrt{n} - 1$. The user sends q_i to the master.
- `Resp(`D, q_i`)` $\rightarrow r_i$. The master computes the PIR, $r_i = \oplus_{t \in S^*} D[t]$, where S^* is the query set received from the user. The master sends r_i back to the user.
- `Recov(`h, r_i`)` $\rightarrow d_i$. Using the parity P_S obtained during the `Query` step, the user recovers the real desired query item $D[i]$ by computing $d_i = P_S \oplus r_i$.

To ensure security against a semi-honest online server, the CK protocol incorporates a cheap *refresh* operation after each query. This operation involves the user generating a new set with $\sqrt{n}-1$ new random keys and obtaining the corresponding parity P_{new} from the replica. The user then adds the previously queried key k to the new set and updates the parity by XORing the key's parity with P_{new}. This refresh operation prevents the server from deducing search access patterns by comparing sets in multiple queries.

Accelerating PIR with BINDEX. The PIR process in the CK protocol effectively conceals search metadata and safeguards user privacy from a semi-honest server. However, a significant drawback is that the CK protocol requires linear time complexity during the membership test (i.e., the `Query` step) and database search step (i.e., the `Resp`) step.

We observe that the core principle of the CK protocol, which leverages the reflexivity of XOR operations on boolean information, can be seamlessly incorporated into BIRDB due to their shared basis in boolean information retrieval. By leveraging this common concept, we can construct efficient (logarithmic complexity and compact) encrypted search indexes on the keys, facilitating rapid membership tests and database searches; further, we can employ the CK protocol to the values, enabling the PIR process to conceal search metadata. In this way, BIRDB simutaneously enables high-performance and metadata-hiding encrypted search in a unified manner.

The end-to-end query workflow operates in the following manner. After constructing the BINDEX on the user's side, the

user submits the chosen OWFs and salted values using b-bit one-time pads (same as the parity) to the BIRDB replica. These OWFs are then used by the replica to construct Bloom filters for each random subset during the `Prep` step; those salted values can later be recovered by the user in the `Recov` step by XORing the corresponding salts owing to the reflexivity property. By doing so, we facilitate efficient membership tests in the subsequent `Query` step, allowing users to quickly determine whether a key is present in a set or not with a complexity of $\mathcal{O}(1)$. In the `Resp` step, the BIRDB master employs the compacted BINDEX to search and retrieve items (Algorithm 1 and Algorithm 2). This operation has a logarithmic time complexity and ensures compactness to reduce I/O costs. Subsequently, the user locally recovers the results and obtains the desired item.

V. PERFORMANCE AND SECURITY ANALYSIS

A. Proof Sketch of Security

BIRDB's security is established through two key lemmas: **Lemma 1** demonstrates the impossibility for an adversary $\mathcal{A}$ to access plaintexts, while **Lemma 2** proves $\mathcal{A}$'s inability to detect the metadata of the specific item being searched. These lemmas collectively form the proof for **Theorem 1**, affirming BIRDB's security.

Lemma 1 (data confidentiality)**.** Given an item with its key (minifilter) mf, value v, and a set of cryptographic hash functions H, it is infeasible for both the BIRDB master and replica to find any input x such that $H(x) = mf$ and reverse the mapping to obtain v.

Proof. We prove this lemma from three perspectives. Firstly, the key (minifilter), as defined in §IV-A, guarantees the irreversibility of plaintext. This is achieved through the use of one-way functions (OWFs) [13], [26] (i.e., H), which provide one-wayness, making it computationally infeasible for any adversary to reverse the transformation of OWF-encoded data. Additionally, v is obfuscated through a one-time pad that is known only to the client, ensuring its irreversibility (§IV-C).

For the second aspect regarding the semi-honest master and replica, the security against the replica is ensured by its lack of knowledge regarding the specific items being queried. The replica remains unaware of the queried items, safeguarding the confidentiality of the search process (§IV-C). Thirdly, security against the master relies on two crucial factors. Firstly, the user's queried subset is uniformly random and remains unknown to the master. This prevents the master from obtaining any information about the user's specific query. Secondly, it is assumed that the two servers involved in the process do not collude (§III-B). By combining these properties, neither the master nor the replica can discover any input x satisfying $H(x) = rf$ and successfully reverse the plaintext of v.

Lemma 2 (metadata confidentiality)**.** Assuming the semi-honest behavior of the master and replica, and their non-collusion, BIRDB ensures complete concealment of specific items that a user fetches.

Proof. **Lemma 2** directly follows from the implementation of the underlying CK protocol within the two-server Private Information Retrieval (PIR) setting [25].

B. High performance

Vanilla BINDEX. In the original uncompressed version of BINDEX, we can achieve a time complexity of $\mathcal{O}(\log N)$ for encrypted search. The inner comparison of minifilter nodes takes $\mathcal{O}(\log N)$ time, where N is the total number of records. The leaf node comparison takes $\mathcal{O}(m)$ time, where m is the bit length of the minifilters. Therefore, the overall time complexity is $\mathcal{O}(\log n)$+$\mathcal{O}(m)$. Since $m \ll n$, the actual time complexity of the uncompressed BINDEX is $\mathcal{O}(\log n)$.

Compressed BINDEX. For the compact version of BINDEX, the average-case time complexity worsens to $\mathcal{O}(\log^2 N)$. This is because the original BINDEX directly computes the positions using OWFs, achieving $\mathcal{O}(1)$ time complexity for each minifilter testing. However, in the compressed version of BINDEX, each minifilter requires a logarithmic search among the segments (or linear models), resulting in $\mathcal{O}(\log n)$ time complexity for each minifilter node's membership test. Hence, the overall time complexity for the compacted BINDEX is $\mathcal{O}(\log^2 N)$.

VI. EVALUATION

Testbed. The experiments were conducted on a cluster of lab machines, each featuring a 2.60GHz Intel E5-2690 V3 CPU, 64GB memory, 40Gbps NIC, and 24 cores. All nodes, including users, master, and the replica, were executed within Docker containers. The average ping latency between the nodes was set at 0.17ms with the aid of Linux traffic control [12].

Baseline. We utilized WiredTiger [41], a prominent NoSQL Key-Value storage engine, to implement all of our baseline systems. WiredTiger is widely employed in industrial MongoDB deployments, making it an appropriate default choice. Our comparative analysis involved three state-of-the-art baseline approaches: HElibDB [24], an imperative oblivious database developed by IBM, which employs Fermat's Little Theorem for comparison on homomorphic encrypted data. However, each query in HElibDB necessitates processing the entire database for metadata protection. We also incorporated the CK [25] two-server OO-PIR protocol, known for its exceptional performance. The CK protocol moves query-independent computation to an offline preprocessing phase. Lastly, we assessed a version of BIRDB without compression, denoted as UC-BIRDB in subsequent evaluations. Although UC-BIRDB maintains logarithmic search complexity through our implementation of a boolean information-based index structure (§IV-A), it exhibits a significantly larger size due to lack of compression.

Workload and default setting. We adopted the same NoSQL key-value workload as CK OO-PIR. This involved generating a range of keys and values with fixed sizes, specifically ranging from 2^{10} to 2^{20}. The key size was set at 8 bytes, while the value size was set at 64 bytes. Unless specified for the scalability study, our evaluations were conducted on a default dataset of 2^{20} key-value pairs. We used point lookup queries for experiments. Each experiment was executed for a duration of 60 seconds, and the results were collected during the middle 30 seconds (i.e., 15s to 45s) to mitigate any potential disturbances caused by system start-up or cool-down. Our extensive evaluation focused on addressing the following research questions:

BIRDB's results (in milliseconds)						
EDB size (# = 2^x)	**10**	**12**	**14**	**16**	**18**	**20**
P0: Prep (Replica)	0.23	0.38	0.6	0.69	1.07	1.41
P1: Query (User)	0.12	0.81	1.26	2.87	3.47	4.6
P2: Refresh (User & Replica)	0.84	1.36	1.55	2.29	3.91	5.44
P3: Response (Master)	0.17	0.22	0.51	0.86	1.12	1.65
P4: Recover (User)	0.04	0.04	0.04	0.04	0.04	0.04
End-to-end ($\sum_1^4 P_i$)	1.17	2.43	3.36	6.06	8.54	11.74

TABLE II: End-to-end latency breakdown of BIRDB.

OO-PIR's results (in milliseconds)						
EDB size (# = 2^x)	**10**	**12**	**14**	**16**	**18**	**20**
P0: Prep (Replica)	0.11	0.27	0.49	0.56	0.89	1.3
P1: Query (User)	2.5	2.41	3.93	4.87	6.54	7.81
P2: Refresh (User & Replica)	0.82	1.35	1.51	2.23	3.8	5.34
P3: Response (Master)	1.45	2.81	3.48	4.64	5.31	6.25
P4: Recover (User)	0.04	0.04	0.04	0.04	0.04	0.04
End-to-end ($\sum_1^4 P_i$)	4.81	6.61	8.96	11.78	15.69	19.45

TABLE III: End-to-end latency breakdown of CK's OO-PIR.

- How efficient is BIRDB compared to baselines?
- What is the communication cost of BIRDB?
- Can BIRDB scale efficiently?
- How sensitive is BIRDB to compression error bound?

A. End-to-end Performance

We evaluated the end-to-end query performance of BIRDB and baselines. We measured latency as the user-perceived response time, while throughput was quantified at the server side (specifically, the BIRDB master). The results, as illustrated in Figure 8, demonstrated that BIRDB exhibited a mean latency that was 5.29 times lower than that of HElibDB. This notable difference can be attributed to BIRDB's new encrypted search approach of BINDEX, which enables logarithmic search complexity. In contrast, HElibDB necessitates linear scans to search for equivalent encrypted data, resulting in significantly longer latency for database queries.

Furthermore, BIRDB achieved a mean latency that was 1.57 times lower than that of OO-PIR. This performance advantage can be attributed to two key factors. Firstly, BIRDB employs Bloom filters during the `Query` step, resulting in a time complexity of $\mathcal{O}(1)$ for membership tests (§IV-C). In contrast, OO-PIR requires $\mathcal{O}(n)$ linear scans of the entire database items. Although this moderately increases BIRDB's communication costs (as shown in Figure 9), it allows BIRDB to achieve higher efficiency along the performance-critical path of the database. Secondly, in the `Response` step, BIRDB's BINDEX reduces the search complexity of OO-PIR from $\mathcal{O}(n)$ to $\mathcal{O}(\log n)$, further contributing to its superior performance.

In terms of query throughput, BIRDB achieved up to 45 times higher throughput than baselines. Furthermore, BIRDB achieved a 1.17 times higher throughput than UC-BIRDB, primarily due to its significantly reduced storage consumption and fewer costly I/O operations to secondary storage. This empirical result aligns with our theoretical evaluation (§III-C), which suggests that achieving both low search complexity and compact indexing is essential for achieving high performance in the encrypted search domain.

Breakdown. To gain insights into the low latency and moderate overhead of BIRDB, we meticulously recorded the time required for each step of the query workflows in BIRDB and OO-PIR. The breakdown of these timings can be found in Table II for BIRDB and Table III for OO-PIR.

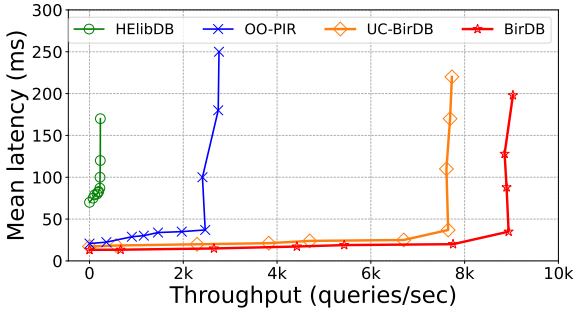
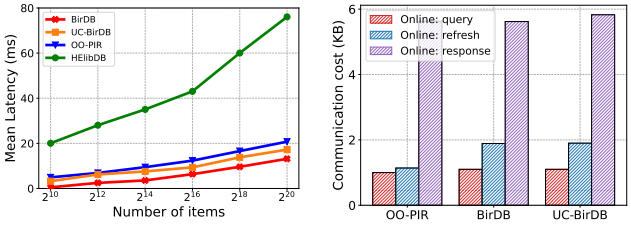


Fig. 8: Throughput and latency comparison among different metadata-hiding baselines, each equipped with 2^{20} items.



(a) Scalability on datasets (b) Communication costs

Fig. 9: (a) Scalability evaluation with varying dataset sizes on all metadata-hiding baselines. (b) Online communication costs for CK OO-PIR, BIRDB, and uncompressed BIRDB.

The latency of both BIRDB and OO-PIR increases as the size of the database grows. However, BIRDB incurs additional preprocessing overhead due to the construction of filters using OWF computations. Fortunately, this overhead is not on the critical path. For both BIRDB and OO-PIR, the online query process consists of two parts: querying an item to the online server and simultaneously refreshing the used set to the offline server. The logic applied by both servers is identical, involving a sublinear number of XOR operations relative to the database size, resulting in low costs. In the refresh step, BIRDB incurs slightly higher latency than OO-PIR due to the reconstruction of filters to achieve $\mathcal{O}(1)$ complexity in the `Query` step. The `Query` and `Response` steps are two additional time-consuming stages in the overall encrypted search process, and BIRDB significantly reduces the associated overhead compared to OO-PIR. This reduction is the primary reason for BIRDB's higher performance.

In summary, BIRDB outperformed all baseline approaches with up to 5.29 times lower latency and up to 45 times higher throughput. It achieved these improvements while maintaining metadata-hiding capabilities, same as all baselines. The communication costs of BIRDB were moderately higher (around 1.65 times) compared to the most efficient baseline (OO-PIR). However, considering the gains in latency and throughput, these increased communication costs were deemed acceptable. Additionally, we observed that achieving both low search complexity and compact indexing is crucial for achieving high performance in the encrypted search domain. While this concept is commonly practiced in industrial database development, to the best of our knowledge, we are the first to propose and validate it in the context of security.

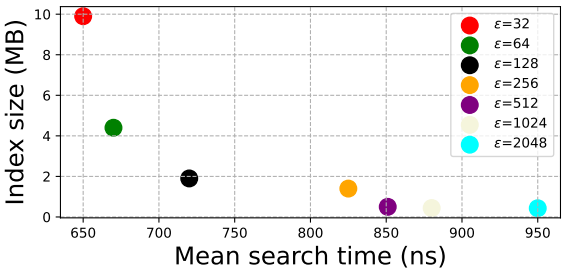


Fig. 10: Sensitivity analysis of compressing BIRDB using different PGM error bounds ϵ.

B. Scalability

Scale to larger datasets. We conducted evaluations of the baselines using key-value datasets of varying sizes, ranging from 2^{10} to 2^{20}. Figure 9(a) depicts the latency scaling of BIRDB, which exhibits a logarithmic pattern consistent with our complexity analysis discussed in §V-B. As the dataset size increases, both BIRDB and OO-PIR experienced a moderate increase in latency. In contrast, HElibDB demonstrated significantly higher latency, attributable to its requirement to perform linear scans of the entire dataset. In contrast, BIRDB and OO-PIR leverage offline preprocessing to move query-independent computation, as evidenced in Table II and Table III. Furthermore, BIRDB exhibited lower latency compared to UC-BIRDB, confirming the enhanced performance achieved by the compressed encrypted search index, as demonstrated in the end-to-end evaluation presented in Figure 8.

C. Sensitivity

Larger error bound, higher cost. We conducted evaluations to assess the sensitivity of compacting BINDEX with different error bounds (i.e., ϵ). Figure 7 illustrates that as the error bound increases, the size of the index decreases significantly and eventually converges, while the mean search (point lookup) time increases. This behavior can be attributed to the fact that a larger error bound results in fewer segments (linear models) being used to index the offsets derived from the original BINDEX's boolean information (§IV-B). Consequently, more entries in the final leaf layer require a binary search. This explains the reduced index size associated with a larger error bound due to the application of fewer segments. Furthermore, the increased search time is a result of fewer segments, as the final binary search at the leaf layer requires more time due to the larger error bound.

VII. Discussion

A. Limitations

BIRDB has two major limitations. Firstly, the BINDEX is not updateable, which means that when the underlying database changes (e.g., through update queries), it requires significant modifications to the search data structure. This includes changes to the layout of BINDEX, the creation of new linear models for compression, and the generation of new chunks for incremental PIR. Finding a satisfactory solution to address this limitation remains an open problem.

Secondly, BINDEX is designed to support searching on a single attribute, making it unsuitable for databases that require searching across multiple attributes, such as graph databases [14], [15]. While BIRDB can handle disjunctive queries by independently searching on the BINDEX of each attribute, it is unable to handle conjunctive queries without integrating BINDEX with multidimensional indexes like $k-d$ tree [87]. Exploring the integration of BINDEX with multidimensional indexes is an interesting future work.

B. Lessons Learned

Based on our theoretical proof and empirical evaluation, we have reached the conclusion that relying solely on logarithmic search complexity is insufficient for achieving high-performance encrypted search. Instead, an encrypted search index should incorporate a compact design similar to traditional insecure databases, which ensures cost-effective I/O operations. By combining the features of low complexity and compactness, an encrypted database can ultimately achieve high-performance query processing.

VIII. Conclusion

We introduce BIRDB, the first encrypted database that efficiently supports encrypted queries while ensuring provable data and metadata confidentiality. BIRDB is built upon the concept of boolean information retrieval, which enables the construction of a secure and efficient index called BINDEX. BINDEX utilizes data membership tests for achieving logarithmic-complexity encrypted searches and employs linear approximation for compression while preserving searchability. Furthermore, BIRDB seamlessly integrates with a two-phase PIR scheme, ensuring metadata hiding. Extensive evaluations conducted on NoSQL workload demonstrate that BIRDB exhibits high efficiency and scalability compared to existing baselines. We have open-sourced BIRDB, and its source code is available at github.com/ndss24p59/BirDB.

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